

Quantum-Safe Blockchain in Hyperledger Fabric

Shahroz Abbas[✉], Ajmery Sultana[✉], *Member, IEEE*, and Georges Kaddoum[✉], *Senior Member, IEEE*

Abstract—With the advances of quantum computing the security of existing cryptographic frameworks is increasingly at risk. Accordingly, in the present study, we investigate the integration of post-quantum cryptographic algorithms into Hyperledger Fabric, a blockchain framework, to safeguard it against emerging quantum threats. To this end, a modified Cryptogen tool was developed to generate X.509 certificates with both classical and post-quantum cryptographic keys. Furthermore, using tools like Hyperledger Caliper and Prometheus for empirical analysis, we demonstrate that this hybrid approach effectively strengthens security without affecting system performance. These results not only improve the security of Hyperledger Fabric, but also offer a practical guide for adding post-quantum cryptography to blockchain technologies.

Index Terms—Blockchain, post-quantum cryptography, hyperledger fabric, hyperledger caliper, prometheus, liboqs, liboqs-go.

I. INTRODUCTION

RECENT advances in quantum computing technology give rise to unprecedented computational power capable of solving complex problems beyond the reach of classical computers. However, this technological quantum leap also poses a giant threat to cryptography algorithms that support modern digital security and protocols. Most of the digital security and protocols are still built around classical cryptography algorithms such as Rivest-Shamir-Adleman (RSA) [1] and Elliptic Curve Digital Signature Algorithm (ECDSA) [2]. On the other hand, quantum computing technology uses quantum mechanical principles to solve these complex problems exponentially faster than classical methods, thereby undermining security of current cryptography algorithms.

Recognizing potential threats posed by quantum computing to classical cryptography systems, the National Institute of Standards and Technology (NIST) is actively engaged in the development and standardization [3] of cryptographic algorithms designed to withstand attacks by quantum computers. More specifically, the NIST has launched large-scale initiatives towards a post-quantum information age that ensures security and integrity.

Most security protocols, including blockchain technology [4], use classical cryptography which is particularly vulnerable. Accordingly, blockchain frameworks like Hyperledger Fabric which are built around classical

cryptography such as RSA [1] or ECDSA [2], are potentially vulnerable to quantum attacks [5].

Hyperledger Fabric [6], an open-source, permissioned blockchain platform designed for enterprise use, provides full modularity, scalability, and security, thus forming a robust base for the development of blockchain applications. Hyperledger Fabric has a different way of implementing smart contracts, and that is what it terms as chaincode [7]. Unlike Ethereum or Bitcoin, Fabric does not rely on Proof-of-Work or Proof-of-Stake for consensus; instead, it uses pluggable consensus algorithms like Practical Byzantine Fault Tolerance or Kafka/raft, to ensure transaction validation and agreement among peers. Hyperledger Fabric is widely used across various industries, including but not limited to finance [8], supply chain [9], healthcare [10], to fulfill various enterprise needs.

Overall, a framework like Fabric uses an x509 certificate to manage identities and authenticate them. At present, the x509 certificates in Hyperledger Fabric use a classical cryptographic algorithm ECDSA [2] that is vulnerable to quantum attacks [5]. Recognizing this vulnerability, in the present study, we integrated quantum-resistant algorithms on our fabric x509 certificates that increase their resistance level against potential quantum attacks and making the Fabric quantum-safe. For the subsequent analysis of the performance of the system throughput on the Post-Quantum Cryptography (PQC), we used Hyperledger Caliper [11] and Prometheus [12].

The main contributions of this letter are:

- *Enhanced Security*: We integrated quantum-resistant algorithms into Hyperledger Fabric's x509 certificates, improving its resistance to quantum attacks.
- *Performance Analysis*: We assessed the performance impact of these algorithms using Hyperledger Caliper and Prometheus, providing a detailed throughput analysis.
- *Future-Proofing*: This integration ensures that Hyperledger Fabric remains resilient to quantum computing advancements.
- *Guidance for Adoption*: Our methodology offers a roadmap for adopting post-quantum cryptographic algorithms in blockchain technologies, aiding future implementations.

The remainder of this letter is organized as follows. Section II describes related work in the field of quantum cryptography and Hyperledger Fabric. In Section III, we provide an overview of the background. Section IV describes the methodology used. In Section V, we report the experimental results. Section VI concludes.

II. RELATED WORK

In this section, we review related work on blockchain technology and quantum encryption. The work in [13] pointed

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Shahroz Abbas and Ajmery Sultana are with the School of Computer Science and Technology, Algoma University, Brampton, ON L6V 1A3, Canada (e-mail: sabbas@algomau.ca; ajmery.sultana@algomau.ca).

Georges Kaddoum is with the Department of Electrical Engineering, Ecole de Technologie Supérieure, Université du Québec, Montreal, QC H3C 1K3, Canada (e-mail: georges.kaddoum@etsmtl.ca).

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out the huge threat that quantum computers pose to classical cryptographic protocols applied in blockchain technology. The authors highlighted how Shor's [14] and Grover's algorithms [15] could potentially compromise these cryptographic systems. A comprehensive review of post-quantum cryptosystems and their application to blockchain has been conducted in [16] where the authors evaluated the most relevant post-quantum blockchain systems and identified their main challenges. Building on this evidence, this letter [17] proposed a method for transitioning towards post-quantum cryptography using XMSS (eXtended Merkle Signature Scheme). The authors highlighted several obstacles, such as key size and processing time, which is one of the major issues affecting the performance of system communication by putting the overhead on security protocols during communication. Furthermore, this letter [18] proposed integrating qTesla, a NIST Round 2 post-quantum signature candidate, into Hyperledger Fabric. However, the authors noted that the large key size of qTesla could potentially generate a large-scale impact on system performance. The authors in [19] advanced Fabric's transition to post-quantum security by modifying its cryptogen binary library. More specifically, the authors implemented signature cryptography such as Dilithium and Falcon. Based upon these findings and taking into account the need for transitioning security protocols towards quantum security, in this letter, we enhanced security protocols of the Hyperledger Fabric towards post-quantum, concurrently maintaining the key size and overhead. Contrary to the approach previously described in [19] based on the outdated Fabric 1.4, that lacked support for current dependencies and relied on digital signatures like Dilithium and Falcon, in the present study, we successfully modernized Fabric 2.5, the latest version, to achieve post-quantum resilience. Our enhancement includes the adoption of a Key Encapsulation Mechanism (KEM) such as Kyber512 and Kyber768 for key exchange, offering faster cryptographic operations (see fig. 7, 8, 9 and 10).

III. BACKGROUND

In this section, we provide an overview of the techniques used in our proposed method, including Hyperledger Fabric, Liboqs-Go, Cryptogen, x509 Certificate, and Public Key Interface (PKI).

A. Hyperledger Fabric

Hyperledger Fabric is a permission-ed blockchain framework; accordingly, access to the network is restricted. Hyperledger Fabric was built by IBM and later taken over under the umbrella of the Hyperledger Project, hosted by Linux Foundation [20]. Fabric supports module architecture as shown in fig. 1.

The peers inside the hyperledger fabric are server nodes that have two components: the World State and Blockchain. As shown in fig. 1, the Chaincode, which is the business logic, is installed into the peers. The World State is a key-value store representing the current state of the ledger. It provides a snapshot of the most recent data and values as updated by committed transactions. The blockchain is the immutable

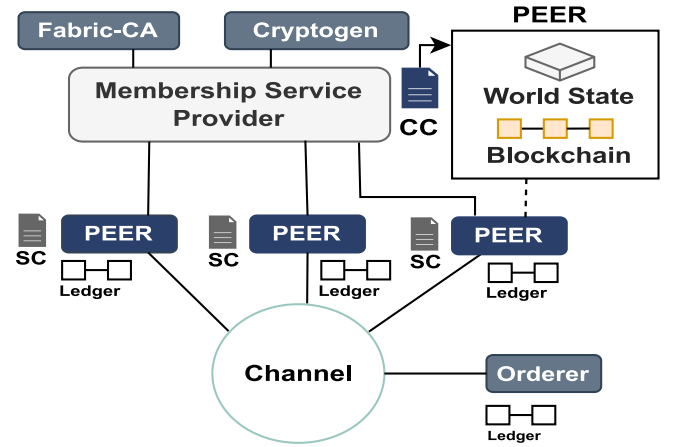


Fig. 1. Fabric Architecture.

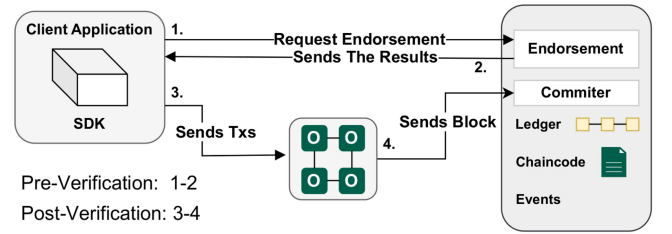


Fig. 2. Transaction Flow.

transaction log. It records all changes that have resulted in the current world state. It is a sequence of interconnected blocks where each block contains separate transactions. These peers are also used for the endorsement of transactions. Endorsement peers are used to validate transaction. Endorsement peers are defined inside the configuration file of the Hyperledger Fabric.

The Membership Service Provider (MSP) inside the Fabric maintains the identity of the Peer. These identities are issued by the Certificate Authority (CA) or Cryptogen tool in the form of an x509 certificate. Without being registered inside the MSP, Peer can not communicate over the channel. Channels inside the Hyperledger Fabric works as dedicated communication pathways within the blockchain network. They allow participants to establish private and secure networks for communication within the broader blockchain, enabling confidential transactions and data sharing among selected participants. The orderer inside Hyperledger Fabric is used to order all incoming transactions after they have received the endorsement. It then aggregates these transactions until the block reaches the specified block size according to the size specified inside the configuration file of the Fabric, creates the block, and adds all transactions inside it. Once the block is formed, it is sent to all peers for subsequent validation and commitment to their ledgers.

The transaction flow is shown in fig. 2 where pre-verification is a very important part of Hyperledger Fabric. Ensuring that transactions are valid, pre-verification prevents malicious actors from spreading a non-valid or unauthorized transaction ahead of its processing by the ordering service for block inclusion. This step helps to reduce computational overheads at the time of consensus and increases overall

efficiency in transaction processing. It is used to validate transactions. Post-Verification, as shown in fig. 2, ensures that the transactions included in a block meet all necessary validation criteria, thereby upholding integrity and reliability of the Hyperledger Fabric ledger.

B. Liboqs-Go

Liboqs-go [21] is a Go (Golang) wrapper for the liboqs library, which provides implementations of quantum-resistant cryptographic algorithms. This wrapper enables Go developers to easily integrate post-quantum cryptographic mechanisms into their applications, using the functionality provided by liboqs [22].

C. Cryptogen

Cryptogen is an external tool inside the Hyperledger Fabric used to generate certificates and keys for different entities in a blockchain network. It is primarily used to create necessary cryptographic artifacts required to set up a Fabric network. Cryptogen is located inside binaries of the Hyperledger Fabric.

D. PKI

The PKI provides a framework for secure communication, identity verification, and transaction integrity in Hyperledger Fabric. PKI is used for secure communication and authentication by managing digital keys and certificates for network entities.

E. x509 Certificate

X.509 is a standard defining the format of public key certificates. X.509 certificates are used in Hyperledger Fabric for identifying and authenticating users. X.509 is a part of the PKI, which is a system for managing digital certificates and public-key encryption.

IV. METHODOLOGY

Hyperledger Fabric uses the Cryptogen tool to generate X.509 certificates and integrate cryptographic material within the network. Cryptogen is an external component of Hyperledger Fabric and can be replaced with a customized version. By generating a new modified Cryptogen for Fabric and replacing the default version found within the Fabric binaries, the cryptographic framework can be enhanced.

At present, X.509 certificates require at least one classical cryptographic algorithm to establish the certificate. To enhance security and make the certificates resistant to quantum attacks [5], a hybrid approach is employed. This involves using both classical and post-quantum keys within the X.509 certificates, thereby upgrading the existing security framework of Hyperledger Fabric. The modified Cryptogen generates X.509 certificates using a combination of classical and post-quantum cryptographic keys. The certificate generation process is shown in fig. 3.

The key points of the process are described below.

Key Generation: The modified Cryptogen tool generates cryptographic keys using both classical algorithms and post-quantum algorithms;

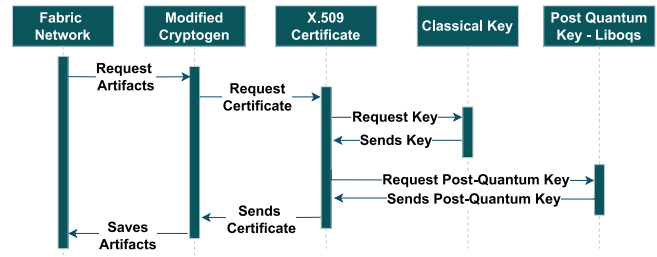


Fig. 3. Certificate Generation Process.

Integration with Liboqs-Go: The post-quantum keys are generated using Liboqs-Go [21], which is a Go wrapper for the liboqs C library [22]. This integration allows for a seamless incorporation of various post-quantum cryptographic (PQC) algorithms, such as Kyber or Dilithium, into the certificate generation process. Using Liboqs-Go, the modified Cryptogen can handle a wide range of PQC algorithms.

Certificate Creation: The generated keys, both classical and post-quantum, are used to create X.509 certificates. These certificates embed the cryptographic keys and are used to authenticate and secure communications within the Hyperledger Fabric network.

Quantum Resistance: By incorporating post-quantum cryptography (PQC) into the certificates, the modified Cryptogen provides an additional security layer. This hybrid approach ensures that certificates are resistant to potential quantum attacks, thus addressing the growing concern of quantum computing's impact on cryptographic security.

Using post-quantum cryptography (PQC), we added another security layer over the Certificate to make the Certificate quantum resistant.

V. EXPERIMENTAL RESULTS

In order to benchmark Hyperledger Fabric, we used Hyperledger Caliper [11]. Caliper is an open-source benchmarking tool designed to measure and compare the performance of different blockchain technologies [23], [24], [25]. It is used to measure various performance metrics such as transaction throughput, latency, and scalability. The process involved integrating the peers' keys into the Caliper network configuration file, defining the schema for the tests, and initiating the benchmarking process on the network. We also used Prometheus [12] to analyze system usage. Prometheus is used to monitor system performance metrics such as CPU utilization, memory usage, and disk input/output (I/O) operations [26]. Furthermore, we used Kyber-512 and Kyber-768 for key exchange, as to compare with the existing post-quantum schemes such as Dilithium-2 and Falcon-512.

Fig. 4 illustrates certificate sizes used to generate artifacts for Hyperledger Fabric. These certificates were used for key exchange processes to authenticate and secure communications within the network. Larger certificates lead to slower communication, as they require more data to be transmitted and processed. As can be seen in Fig. 4, Kyber-512 and Kyber-768 have lower certificate sizes as compared to other post-quantum schemes.

ALGORITHMS	CERT SIZE (BYTES)
ECDSA	867 Bytes
KYBER-512	1,988 Bytes
KYBER-768	2,508 Bytes
NTRUENCRYPT	2,472 Bytes
FALCON-512	3,044 Bytes
FALCON-1024	5,092 Bytes
DILITHIUM-2	5,990 Bytes
DILITHIUM-3	8,027 Bytes

Fig. 4. Certificate Size.

ALGORITHMS	KEY SIZE (BYTES)
ECDSA	91 Bytes
KYBER-512	800 Bytes
KYBER-768	1,184 Bytes
NTRUENCRYPT	1,158 Bytes
FALCON-512	897 Bytes
FALCON-1024	1,783 Bytes
DILITHIUM-2	1,312 Bytes
DILITHIUM-3	1,952 Bytes

Fig. 5. Key Size.

ALGORITHMS	KEY GENERATION TIME (SECONDS)
ECDSA	0.00156 S
KYBER-512	0.00253 S
KYBER-768	0.00280 S
NTRUENCRYPT	0.00325 S
FALCON-512	0.02280 S
FALCON-1024	0.15720 S
DILITHIUM-2	0.00370 S
DILITHIUM-3	0.00350 S

Fig. 6. Key Generation Time.

Fig. 5 illustrates key sizes of algorithms used for generating artifacts for Hyperledger Fabric and compares key lengths of classical algorithms and post-quantum algorithms. The larger key sizes leads to increased computational overhead and longer processing times. As can be seen in Fig. 5, Kyber-512 and Kyber-768 have lower key sizes as compared to other post-quantum schemes.

Fig. 6 illustrates the key generation time of algorithms used to generate artifacts for Hyperledger Fabric. Key generation time refers to the time cryptographic algorithms need to produce keys used for securing transactions and communications within the network. As shown in Fig. 6, Kyber-512 and Kyber-768 have lower key generation times as compared to the other post-quantum schemes.

Fig. 7 demonstrates the system throughput of Hyperledger Fabric when using both classical and post-quantum cryptographic approaches. System throughput refers to the number of transactions processed by the network within a given time frame, indicating the overall performance and capacity of the system. As can be seen Fig. 7, Kyber-512 and Kyber-768 have higher throughput as compared to the other post-quantum schemes.

Fig. 8 shows the system latency of Hyperledger Fabric when using both classical and post-quantum cryptographic

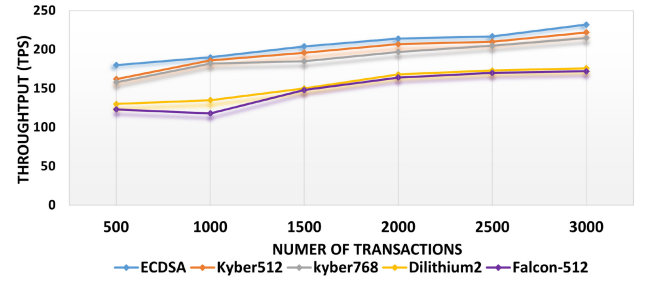


Fig. 7. Throughput.

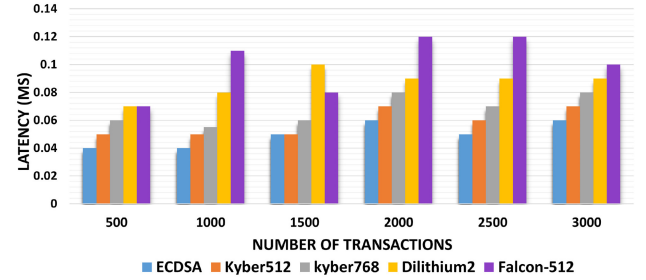


Fig. 8. Latency.

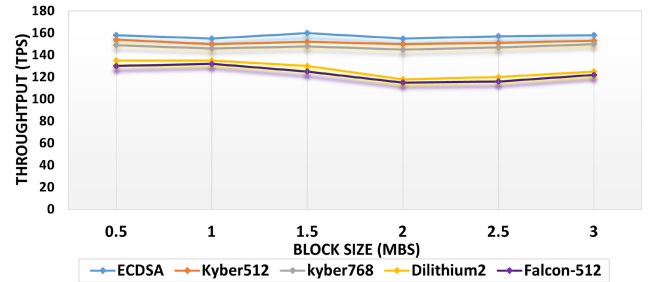


Fig. 9. Block Size.

approaches. System latency refers to the time delay between the submission of a transaction and its final confirmation on the blockchain. As can be seen in Fig. 8, Kyber-512 and Kyber-768 consume less latency as compared to the other post-quantum schemes.

Fig. 9 illustrates the block size of Hyperledger Fabric when using both classical and post-quantum cryptographic approaches. Block size refers to the maximum amount of data or number of transactions contained in a single block (see Section III for further discussion). As shown in Fig. 9, Kyber-512 and Kyber-768 yield higher throughput as compared to the other post-quantum schemes.

Finally, Fig. 10 shows system usage metrics of Hyperledger Fabric when using both classical and post-quantum cryptographic approaches. The results reveal that Kyber-512 and Kyber-768 consume less system usage as compared to the other post-quantum schemes.

VI. CONCLUSION

Quantum computing pose serious challenges to the security of classical cryptographic systems. In this letter, we addressed these challenges via integrating post-quantum cryptographic algorithms into Hyperledger Fabric, the enterprise

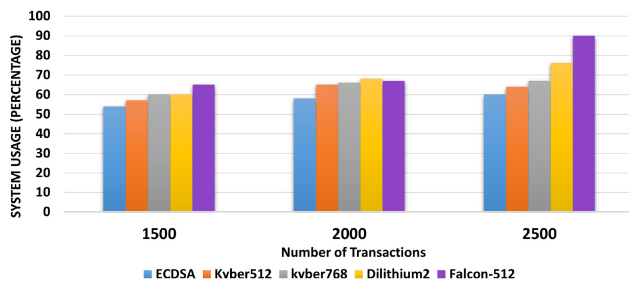


Fig. 10. System Usage.

blockchain framework. We developed a modified Cryptogen tool that generates hybrid X.509 certificates with both classical and post-quantum cryptographic keys. Hence, leading to a quantum-safe blockchain. To ensure the effectiveness of this approach, We have performed an in-depth analysis using Hyperledger Caliper and Prometheus. The empirical results show that post-quantum cryptography can be integrated into a blockchain framework for protection against quantum attacks without unnecessarily compromising system performance. This trade-off between improved security and better operational efficiency is very essential in safeguarding blockchain technology for the future.

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